



**RIGA NORDIC
UNIVERSITY**

Studiju programma

4220 Informācijas sistēmas (bak.)

Dabaszinātņu un datortehnoloģiju katedra

**RĪGAS ZIEMEĻVALSTU UNIVERSITĀTE
(RNU)**

**Informācijas sistēmu integrācijas brieduma novērtējums
projektu vadības birojos: pierādījumi no Latvijas uzņēmumiem**

BAKALaura DARBS

Students:

Amina Showkath

Darba vadītājs:

Dr.sc.ing. prof. Andrejs Bondarenko

RĪGA 2026

Study Programme

4220 Information Systems (BSc)

Department of Natural Sciences and Computer Engineering

**RIGA NORDIC UNIVERSITY
(RNU)**

**Assessment of Information System Integration Maturity in
Project Management Offices: Evidence from Latvian
Enterprises**

BACHELOR'S THESIS

Student: Amina Showkath

Supervisor: Dr.sc.ing. prof. Andrejs Bondarenko

Riga 2026

Anotācija

Šis bakalaura darbs pētīja informācijas sistēmu integrācijas briedumu projektu vadības biroju (PVB) vidē Latvijas vidējos un lielos uzņēmumos. Tā kā organizācijas arvien vairāk paļaujas uz vairākiem digitālajiem rīkiem projektu plānošanai, pārskatošanai un pārvaldībai, sistēmu integrācijas pakāpei ir tieša ietekme uz datu kvalitāti, pārskatu efektivitāti un pārvaldības standartizāciju. Neskatoties uz tās praktisko nozīmi, šī tēma Latvijas kontekstā bija saņēmusi ierobežotu empīrisku uzmanību.

Bakalaura darbs sastāv no piecām nodaļām. 1. nodaļā aplūkoti teorētiskie pamati, definējot galvenos jēdzienus, piemēram, PVB, informācijas sistēmu integrāciju, brieduma modeļus, biznesa inteliģenci un datu kvalitāti, kā arī apskatīti esošie pētījumi par PVB digitalizāciju un integrācijas brieduma ietvariem. 2. nodaļā sniegta pētījuma metodoloģija, iekļaujot četru līmeņu Integrācijas Brieduma Modeļa izstrādi un strukturēta aptaujas instrumenta veidošanu. 3. nodaļā veikts empīriskais pētījums, prezentējot aptaujas rezultātus, kas iegūti no Latvijas uzņēmumu pārstāvjiem ar formālām PVB struktūrām, papildināts ar vienu strukturētu ekspertu interviju un statistisko analīzi. 4. nodaļā rezultāti tika interpretēti esošās literatūras kontekstā un izstrādāti pieci praktiski ieteikumi informācijas sistēmu integrācijas brieduma uzlabošanai PVB vidē. 5. nodaļā aplūkoti nākotnes pētījumu virzieni un plašākas sociālās un ekonomiskās sekas.

Pētījuma metodoloģiskais pamats ietver sistemātisku literatūras apskatu, aptaujas bāzētu kvantitatīvo pētījumu, vienu strukturētu ekspertu interviju, aprakstošo statistisko analīzi, kā arī Spīrmena ranga korelācijas analīzi.

Pētījums atklāja, ka 70% PVB vižu Latvijā darbojas vidējā integrācijas brieduma līmenī (2. vai 3. līmenis), un ka augstāks integrācijas briedums ir statistiski nozīmīgi saistīts ar lielāku pārskatu automatizāciju ($r=0,405$, $p=0,010$), labāku datu kvalitāti ($r=0,321$, $p=0,043$) un stiprāku pārvaldības standartizāciju ($r=0,403$, $p=0,010$). Visi četri pētījuma hipotēzi tika apstiprināti. Rezultāti sniedz uz pierādījumiem balstītus norādījumus organizācijām, kas vēlas uzlabot savas projektu vadības informācijas sistēmas.

Bakalaura darbs izstrādāts **57** lapaspusēs un ietver **9** tabulas, **0** attēlus, **0** algoritmus, **0** programmu kodu izdrukas (listings), **2** pielikumus, kā arī literatūras sarakstu ar **12** avotiem.

Atslēgvārdi: Projektu vadības birojs, Informācijas sistēmu integrācija, Integrācijas

brieduma modelis, Pārskatu automatizācija, Datu kvalitāte, Biznesa intelīģence, Pārvaldības standartizācija.

Abstract

This Bachelor's thesis examined the integration maturity of information systems within Project Management Office (PMO) environments in medium and large Latvian enterprises. As organizations increasingly rely on multiple digital tools for project planning, reporting, and governance, the degree to which these systems are integrated has a direct impact on data quality, reporting efficiency, and governance standardization. Despite its practical relevance, this topic had received limited empirical attention in the Latvian context.

The Bachelor Paper consists of five chapters. Chapter 1 reviewed the theoretical background, defining key concepts such as PMO, information system integration, maturity models, business intelligence, and data quality, and surveyed existing research on PMO digitalization and integration maturity frameworks. Chapter 2 presented the research methodology, including the design of a four-level Integration Maturity Model and the development of a structured survey instrument. Chapter 3 contained the empirical study, presenting the results of a survey conducted among representatives of Latvian enterprises with formal PMO structures, supplemented by one structured expert interview and statistical analysis. Chapter 4 discussed the findings in the context of existing literature and proposed five practical recommendations for improving information system integration maturity in PMO environments. Chapter 5 reflected on future research directions and the broader societal and economic implications of the study.

The methodological basis of the study included systematic literature review, survey-based quantitative research, one structured expert interview, descriptive statistical analysis, and Spearman rank correlation analysis.

The study found that 70% of PMO environments in Latvia operate at intermediate integration maturity levels (Levels 2 or 3), and that higher integration maturity is significantly associated with greater reporting automation ($r=0.405$, $p=0.010$), better data quality ($r=0.321$, $p=0.043$), and stronger governance standardization ($r=0.403$, $p=0.010$). All four research hypotheses were supported. The findings provide evidence-based guidance for organizations seeking to improve their project management information systems.

The Bachelor Paper is written in **57** pages and contains **9** tables, **0** figures, **0** algorithms, **0** listings, **2** annexes, and a list of references with **12** sources.

Keywords: Project Management Office, Information System Integration, Integration Maturity Model, Reporting Automation, Data Quality, Business Intelligence, Governance Standardization.

Key words / Atslēgvārdi

English

Project Management Office

Information System Integration

Integration Maturity Model

Reporting Automation

Data Quality

Business Intelligence

Governance Standardization

Latvian

Projektu vadības birojs

Informācijas sistēmu integrācija

Integrācijas brieduma modelis

Pārskatu automatizācija

Datu kvalitāte

Biznesa inteliģence

Pārvaldības standartizācija

Contents

Anotācija	1
Abstract	3
Key words / Atslēgvārdi	5
Introduction	13
1 Literature Review and Theoretical Background	16
1.1 Core Concepts and Definitions	16
1.1.1 Project Management Office (PMO)	16
1.1.2 Project Management Information System (PMIS)	16
1.1.3 Information System Integration	16
1.1.4 Integration Maturity Model	17
1.1.5 Business Intelligence and Dashboard Systems	17
1.1.6 Data Quality	17
1.1.7 Reporting Automation	18
1.1.8 Medium and Large Enterprises in the Latvian Context	18
1.2 Related Work: Themes, Methods, and Evidence	18
1.2.1 Traditional PMO Research and Manual Integration Practices	18
1.2.2 API-Based and Platform-Level Integration Research	19
1.2.3 Business Intelligence and Governance Standardization	19
1.2.4 Maturity Model Frameworks in Information Systems Research	20
1.2.5 Digital Maturity Research in Baltic and Eastern European Enterprises	20
1.3 Theoretical Framework	20
1.3.1 Foundations of the Integration Maturity Model	21
1.3.2 The Four-Level Integration Maturity Model	21
1.3.3 Variables and Relationships	23

1.4	Research Motivation and Direction	23
1.5	Chapter Summary	25
2	Research Methodology	26
2.1	Research Design and Approach	26
2.2	Operationalization of the Integration Maturity Model	26
2.3	Survey Instrument Design	26
2.4	Survey Questionnaire	27
2.5	Expert Interview Design	28
2.6	Data Collection Procedure	28
2.7	Data Analysis Methods	29
2.8	Validity and Reliability	29
2.9	Research Ethics	30
2.10	Limitations of the Research Design	30
2.11	Chapter Summary	31
3	Empirical Study	33
3.1	Sample Characteristics	33
3.1.1	Organization Size	33
3.1.2	Industry Distribution	33
3.1.3	PMO Maturity and Scope	33
3.2	Descriptive Statistics	34
3.2.1	Integration Maturity Level Distribution	34
3.2.2	Descriptive Statistics for Key Constructs	34
3.3	Hypothesis Testing	35
3.3.1	Hypothesis 1	35
3.3.2	Hypothesis 2	35
3.3.3	Hypothesis 3	35
3.3.4	Hypothesis 4	36
3.4	Expert Interview Findings	36
3.4.1	Expert 1 — Project Management Professional	36
3.5	Chapter Summary	37
4	Discussion and Recommendations	38

4.1	Discussion of Findings	38
4.1.1	Integration Maturity Distribution in Latvian PMOs	38
4.1.2	Integration Maturity and Reporting Automation	38
4.1.3	Integration Maturity and Data Quality	39
4.1.4	Integration Maturity and Governance Standardization	39
4.2	Comparison with Existing Literature	39
4.3	Unexpected Findings and Theoretical Implications	40
4.3.1	Unexpected Findings	40
4.3.2	Theoretical Implications	40
4.4	Practical Recommendations	41
4.4.1	Recommendation 1 — Prioritize API-Based Integration	41
4.4.2	Recommendation 2 — Invest in Centralized BI Dashboards	41
4.4.3	Recommendation 3 — Consider Low-Code Integration Platforms	41
4.4.4	Recommendation 4 — Address Local System Interoperability	41
4.4.5	Recommendation 5 — Invest in Data Literacy	42
4.5	Limitations of the Study	42
4.6	Chapter Summary	42
5	Future Work and Broader Reflections	43
5.1	Future Research Directions	43
5.2	Broader Societal and Economic Implications	44
5.3	Chapter Summary	44
	Conclusions	45
	Literature	48
	A Survey Questionnaire	50
	B Expert Evaluation Criteria for Final Thesis Quality	54
	Apliecinājums / Affirmation	55
	Pateicības / Acknowledgments	56

List of Figures

List of Tables

1	Correspondence of research tasks to thesis chapters	14
1.1	Comparison of related research strands	21
1.2	Four-Level Integration Maturity Model for PMO Information Systems . . .	22
2.1	Operationalization of Integration Maturity Model levels	27
2.2	Survey instrument items by construct and hypothesis	32
3.1	Distribution of IS Integration Maturity Levels	34
3.2	Descriptive Statistics for Key Constructs	34
3.3	Summary of Hypothesis Testing Results	36
B.1	Expert evaluation criteria for the quality of the final thesis	54

List of Algorithms

Listings

Introduction

Topicality of the problem: In the context of growing digital transformation, Project Management Offices (PMOs) in Latvian enterprises increasingly rely on multiple information systems for planning, reporting, and governance. However, many organizations struggle with fragmented system landscapes where data must be manually consolidated across tools, leading to inefficiencies, data quality issues, and delayed decision-making. Despite the practical significance of this challenge, empirical research on information system integration maturity in PMO environments within Latvia remains limited.

Aim of the study: The aim of the study is to assess the integration maturity of information systems within PMO environments in medium and large Latvian enterprises and examine its relationship with reporting automation, data quality, and governance standardization.

Object of the study: Information systems within Project Management Office environments in Latvian enterprises.

Subject of the study: The integration maturity of information systems and its relationship with reporting automation, data accuracy, and governance standardization in PMO environments.

Research problem: Many PMOs in Latvian enterprises operate with low levels of information system integration, resulting in poor architectural coherence, manual data consolidation, reduced reporting efficiency, and inconsistent governance practices.

Tasks of the study:

1. Conduct a systematic literature review on PMO digitalization, information system integration architectures, and maturity models relevant to IS environments.
2. Adapt and operationalize an Integration Maturity Model (Levels 1–4) with clearly defined, survey-measurable criteria for each level, based on integration mechanisms, automation level, reporting structure, and BI centralization.
3. Design and validate a structured survey instrument containing Likert-scale and categorical questions aligned with the Integration Maturity Model dimensions.

4. Collect empirical data through distribution of the survey to representatives of medium and large Latvian enterprises (50+ employees) with formal PMO structures, targeting 30–50 responses.
5. Conduct a structured expert interview to supplement survey findings with architectural depth and qualitative insights.
6. Perform descriptive statistical analysis to determine the distribution of integration maturity levels across the sampled PMO environments.
7. Apply correlation analysis (Spearman rank correlation) to examine relationships between integration maturity and reporting automation, data accuracy, and governance standardization.
8. Interpret findings, draw conclusions regarding the hypotheses, and formulate practical recommendations for PMO information system integration improvement in Latvian enterprises.

The correspondence between the research tasks and the thesis chapters is presented in Table 1.

Table 1. Correspondence of research tasks to thesis chapters

Research Task	Thesis Chapter
Task 1 – Literature review on PMO digitalization and maturity models	Chapter 1
Task 2 – Design of Integration Maturity Model (Levels 1–4)	Chapter 1
Task 3 – Survey instrument design and validation	Chapter 2
Task 4 – Empirical data collection from Latvian enterprises	Chapter 3
Task 5 – Expert interview	Chapter 3
Task 6 – Descriptive statistical analysis	Chapter 3
Task 7 – Correlation analysis	Chapter 3
Task 8 – Conclusions and practical recommendations	Chapter 4

Hypotheses:

1. Most PMO environments in medium and large Latvian enterprises operate at intermediate (Levels 2–3) rather than advanced integration maturity levels.

2. PMOs utilizing API-based or platform-level integration demonstrate higher reporting automation than those relying primarily on manual data consolidation.
3. Integration maturity positively correlates with perceived data accuracy and timeliness.
4. The use of centralized BI/dashboard systems is associated with higher governance standardization in PMO processes.

Research methods: The study applies systematic literature review, survey-based quantitative research, a structured expert interview, descriptive statistical analysis, and Spearman rank correlation analysis.

Approbation of the study: The research findings were presented during the Bachelor's thesis defence at Riga Nordic University in 2026.

1. Literature Review and Theoretical Background

1.1. Core Concepts and Definitions

This section defines the foundational terminology used throughout the thesis.

Each term is defined once and used consistently in subsequent chapters.

1.1.1. Project Management Office (PMO)

A Project Management Office (PMO) is a central part of an organization that is in charge of setting and keeping project management standards, governance frameworks and reporting structures across the whole company [11]. According to the Project Management Institute, PMOs can be classified into three types based on their degree of control and influence:

- **Supportive PMO** — provides templates, best practices, and guidance with minimal direct control over projects;
- **Controlling PMO** — enforces compliance with project management frameworks and requires the use of specific methodologies;
- **Directive PMO** — takes direct control of projects, with project managers reporting directly to the PMO.

1.1.2. Project Management Information System (PMIS)

A project management information system (PMIS) is a software-based system or set of tools that helps with planning, doing, tracking, and reporting on projects in a PMO setting. [7].

Some common PMIS tools are Microsoft Project, Jira, SAP, and Power BI. A PMIS can be one big platform or a group of smaller tools that work together, which makes it hard to integrate information systems.

1.1.3. Information System Integration

Integrating information systems means connecting two or more software systems so that data can move between them automatically without any help from people

[6]. In PMO settings, project tracking tools, financial systems, and reporting platforms usually work together. Integration can be as simple as

manually putting together data or as more complex as using Application Programming Interfaces (APIs) or unified platform ecosystems to automatically sync data in real time.

1.1.4. Integration Maturity Model

A maturity model is a structured framework that shows how a capability develops over a set of levels. [10].

A maturity model indicates the progression an organization achieves in information system integration, evolving from elementary, manual practices to comprehensive, automated, platform-level integration. This thesis modifies the idea of maturity modeling applied to the specific situation of PMO information systems, implying a four-level Integration Maturity Model as the fundamental theoretical framework.

1.1.5. Business Intelligence and Dashboard Systems

Business Intelligence (BI) stands for the tools, methods, and ways of doing things used to gather, combine, analyze, and show business data to help making decisions.

[9] In PMO settings, BI systems usually show up as centralized dashboards that bring together data from many different projects and financial systems into one view.

In this thesis, the presence or absence of centralized BI systems is used as an indicator of governance standardization maturity.

1.1.6. Data Quality

Data quality refers to the degree to which data is accurate, complete, consistent, and timely for its intended purpose. [2]

When it comes to PMO reporting, the level of system integration has a direct influence on the quality of the data: Data that is put together manually by humans is more likely to have errors and require more time, while automated integration lowers the chance of

inconsistencies. This thesis assesses data quality via perceived accuracy and timeliness as

indicated by PMO representatives.

1.1.7. Reporting Automation

Reporting automation is the use of software tools and system integrations to automatically generate project status reports, dashboards, and portfolio summaries without manual data entry [11]. Automated reporting reduces the workload on PMO staff and improves the timeliness and consistency of management information. In this thesis, reporting automation refers to the degree to which PMO reports are produced automatically from integrated systems rather than compiled manually.

1.1.8. Medium and Large Enterprises in the Latvian Context

As per the European Commission's classification for company sizes, a medium-sized company is one that has between 50 and 249 employees and a large company employs 250 or more people, [1]. This thesis focuses on medium and large Latvian enterprises due to their higher tendency to have set up formal PMO structures and information that is complex enough system landscapes to make integration maturity a real issue

1.2. Related Work: Themes, Methods, and Evidence

This section surveys existing research on PMO digitalization, information system integration, and maturity models. Prior work is organized into five thematic strands, which are contrasted to identify the gap that the present thesis addresses.

1.2.1. Traditional PMO Research and Manual Integration Practices

Early research on Project Management Offices focused primarily on process standardization, governance frameworks, and the role of PMOs in improving project delivery outcomes [5]. In this body of literature, information systems were treated as secondary tools — typically spreadsheets and standalone reporting software — rather than integrated platforms. Manual data consolidation was the dominant practice, with PMO staff collecting data from multiple sources and compiling reports manually. While this approach established the theoretical foundations of PMO governance, it did not address

the challenges of system fragmentation or the consequences of low integration maturity on reporting efficiency and data quality. The primary limitation of this strand is its failure to account for the growing complexity of modern PMO information system landscapes.

1.2.2. API-Based and Platform-Level Integration Research

A second line of research looks at the technical aspects of information integration of systems, with a focus on API-based connectivity and platform-level ecosystems [6]. Research in this domain illustrates that organizations adopting API-driven integration architectures attain significantly higher levels of reporting automation and data consistency compared to those that depend on manual

or file-based data transfer[7]. Ecosystems for platforms like Native integration features in Microsoft 365, SAP, and Jira cut down on the need for manual

work. This research strand is primarily focused on technical aspects and does not methodically

analyze the organizational or governance effects of integration maturity in PMO-specific situations. Furthermore, implementation costs and the technical expertise required for

API-based integration present significant barriers for medium-sized enterprises.

1.2.3. Business Intelligence and Governance Standardization

A third research strand examines the correlation between Business Intelligence (BI) systems and organizational governance [9]. Studies consistently

show that enterprises with centralized BI dashboards have higher levels of governance standardization, as real-time data visibility encourages better, consistent

decision-making and performance monitoring across projects[11]. This strand is particularly relevant to the present thesis, as it provides theoretical support for Hypothesis 4 — that the use of centralized BI systems is associated with higher governance standardization in PMO processes. The main limitation of this research strand is its focus on large enterprises, with limited attention

to medium-sized organizations or specific regional contexts.

1.2.4. Maturity Model Frameworks in Information Systems Research

The fourth strand covers research on maturity models used in information systems and organizational capabilities [10]. Well-known frameworks such as CMMI, OPM3, and P3M3 help organizations measure and improve their capabilities across defined levels. More recent work has extended these ideas to digital transformation, producing Digital Maturity Models that assess how digital an organization is across multiple dimensions [8]. However, none of these frameworks specifically address IS integration maturity in PMO environments, which is the gap this thesis fills.

1.2.5. Digital Maturity Research in Baltic and Eastern European Enterprises

The fifth strand comprises new studies on digital transformation and maturity in enterprises in the Baltic and Eastern Europe. Research demonstrates that companies in Latvia and the rest of the Baltic region tend to have lower levels of digital maturity compared to Nordic peers, especially in the areas of integrating systems

and automating processes[3]. While some research has examined e-government digitalization and SME digital adoption in Latvia, empirical studies specifically addressing information system integration maturity in PMO environments remain absent from the literature. This represents a significant gap, as the Latvian enterprise context — characterized by a predominance of medium-sized organizations and a rapidly evolving digital landscape — may exhibit distinct integration maturity patterns that differ from those documented in Western European or North American studies. The present thesis directly addresses this gap by providing the first empirical assessment of PMO IS integration maturity in Latvian enterprises.

1.3. Theoretical Framework

This section presents the theoretical lens that grounds the empirical investigation. Based on the review of existing maturity models and IS integration research in Section 1.2, this thesis proposes a four-level Integration Maturity Model (IMM) specifically adapted to PMO information system environments.

Table 1.1. Comparison of related research strands

Strand	Focus	Strength	Limitation
Traditional PMO	Process standardization	Strong governance theory	No IS integration focus
API Integration	Technical connectivity	Automation evidence	No PMO context
BI & Governance	Dashboard systems	Governance link	Large enterprises only
Maturity Models	Capability levels	Structured framework	Not PMO IS specific
Baltic Digital Maturity	Regional context	Latvia relevance	No PMO IS empirics

1.3.1. Foundations of the Integration Maturity Model

The proposed model draws on three theoretical foundations. First, the staged progression concept from the Capability Maturity Model Integration (CMMI) [10], which posits that organizations advance through discrete, measurable levels of capability. Second, the enterprise integration patterns framework [6], which classifies integration mechanisms by their degree of automation and architectural sophistication. Third, the data quality dimensions framework [2], which links integration architecture to data accuracy, completeness, and timeliness outcomes.

1.3.2. The Four-Level Integration Maturity Model

The Integration Maturity Model proposed in this thesis defines four levels of IS integration maturity in PMO environments, as presented in Table 1.2.

Level 1 — Initial.

At this level, PMO environments depend on putting together data by manual means. People usually use spreadsheets to put together reports from project data collected from many different tools that aren't connected to each other. There is no organized way for data to move between systems, which makes it very likely that mistakes, inconsistencies, and de-

Table 1.2. Four-Level Integration Maturity Model for PMO Information Systems

Level	Name	Integration Mechanism	Reporting	Governance
1	Initial	No integration; fully manual data consolidation via spreadsheets	Fully manual; high error rate	Ad hoc; no standardization
2	Developing	File-based integration (CSV/Excel exports); partial automation	Semi-manual scheduled exports	Basic standards; inconsistent
3	Defined	API-based point-to-point integration; automated data flows	Mostly automated; near real-time	Defined processes; mostly consistent
4	Optimized	Platform-level integration; unified data ecosystem with BI	Fully automated; real-time dashboards	fully standardized continuous improvement

lays in

reporting will happen. There are no set rules for governance practices

Level 2 — Developing.

At Level 2, some level of semi-automated, there is integration through file-based data exchange, like when systems transmit CSV or Excel files to each other on a set schedule.

This helps to save some time, but people still need to do a lot of work to combine and check the

data. Reporting is partially automated, and governance standards are only partially defined and

not consistently followed.

Level 3 — Defined. Level 3 organizations have implemented API-based point-to-point integrations between key PMO systems, enabling automated data flows between project tracking, financial, and reporting tools. This enables data flow automatically between tools for tracking projects, finances, and reports. Reporting is mostly done by computers and is almost real-time. Most PMO functions

have clear rules for governance that they always follow

Level 4 — Optimized. At the highest level of maturity, PMO environments operate within a unified platform ecosystem where all systems are fully integrated

and data flows in real time. Centralized BI dashboards give you a full view of your portfolio

, and all governance processes are fully standardized with ways to keep getting better. This level is the standard for how well information systems should work together in PMO settings.

1.3.3. Variables and Relationships

The model defines the following key variables and relationships that will be tested empirically in Chapter 3:

- **Independent variable:** Integration Maturity Level (1–4), operationalized through survey items measuring integration mechanisms, automation level, and BI centralization.
- **Dependent variables:** Reporting automation degree, perceived data accuracy and timeliness, and governance standardization level.
- **Expected relationships:** Higher integration maturity levels are expected to be positively associated with all three dependent variables, consistent with the hypotheses stated in the Introduction.

1.4. Research Motivation and Direction

Problem significance. For Latvian enterprises, the fact that information systems get mixed up in PMO environments is

a major operational problem. When project tracking, financial, and reporting tools don't work

together, the PMO staff has to spend a lot of time manually putting together data, which raises

the chance of mistakes and makes management reports less timely. This lack of efficiency has a

direct impact on how well strategic decisions are made at the portfolio level.

The people who are most affected are PMO managers, project sponsors, and executive leaders

who need accurate and up-to-date project data to decide how to spend money and resources.

Identified gap. As established in Section 1.2, current research on IS integration maturity fails to

specifically focus on PMO environments, and there is a lack of empirical studies investigating

this phenomenon within the Latvian enterprise context. The gap can be stated

precisely as follows: *the distribution of IS integration maturity levels across PMO environments in Latvian enterprises, and its relationship with reporting automation, data quality, and governance standardization, has not been empirically measured or documented.*

This gap is both testable and measurable

through survey-based quantitative research.

Direction and scope. This thesis addresses the gap by conducting an empirical survey-based study of IS

integration maturity in PMO environments in medium and large Latvian businesses.

The research utilizes the four-level Integration Maturity Model established in Section 1.3 as the primary measurement framework.

The scope is intentionally limited to enterprises with 50 or more employees that uphold a

formal Project Management Office (PMO) structure, as smaller organizations are unlikely to

possess sufficiently intricate information systems landscapes to render integration maturity a

significant issue. The study explicitly excludes the analysis of individual project outcomes,

software implementation processes, or the technical architecture of specific integration

solutions. This direction is directly related to the goals and tasks set out in the introduction.

Success criteria. The study will be considered successful if it achieves the following outcomes:

- (1) a valid and reliable survey instrument is developed and administered to at least 30 respondents from Latvian enterprises with formal PMO structures;
- (2) the distribution of integration maturity levels across the sample is described using descriptive statistics; (3) statistically significant relationships between integration maturity and the dependent variables (reporting automation, data quality, governance standardization) are tested using Spearman rank correlation analysis; and (4) practical recommendations for PMO IS integration improvement are formulated based on the empirical findings.

1.5. Chapter Summary

This chapter established the theoretical and conceptual foundations of the thesis. Section 1.1 defined eight key terms — PMO, PMIS, IS integration, integration maturity model, business intelligence, data quality, reporting automation, and medium/large enterprise — providing a consistent conceptual vocabulary for the remainder of the thesis. Section 1.2 surveyed five thematic strands of related research, contrasting traditional PMO research, API integration studies, BI-governance research, maturity model frameworks, and Baltic digital maturity research, and identified a concrete empirical gap. Section 1.3 proposed a four-level Integration Maturity Model as the primary theoretical framework, defining the variables and relationships to be tested empirically. Section 1.4 articulated the problem significance, the measurable research gap, the scope of the study, and the success criteria.

Conclusions for Chapter 1.

The literature review confirms that while substantial research exists on PMO governance, IS integration architectures, and digital maturity models, no empirical study has specifically examined IS integration maturity in PMO environments within Latvia. The proposed four-level Integration Maturity Model provides a theoretically grounded and operationalizable framework for addressing this gap. These foundations motivate the research methodology presented in Chapter 2 and the empirical study conducted in Chapter 3.

2. Research Methodology

This chapter presents the research methodology applied in this thesis. It describes the overall research design, the operationalization of the Integration Maturity Model, the design of the survey instrument and expert interview, data collection procedure, and the statistical methods used to analyze the collected data.

2.1. Research Design and Approach

This thesis adopts a quantitative research design, complemented by a qualitative expert interview. The principal research methodology is a cross-sectional survey study, in which data is gathered from various organizations at a single point in time. This design is appropriate for the research objective, which is to evaluate the present distribution of IS integration maturity levels within PMO environments in Latvian enterprises and to examine its correlations with reporting automation, data quality, and governance standardization [12].

The research follows a deductive approach, in which the four hypotheses stated in the Introduction are derived from the theoretical framework presented in Chapter 1 and tested against empirical data collected from Latvian enterprises.

2.2. Operationalization of the Integration Maturity Model

The four-level Integration Maturity Model (IMM) proposed in Chapter 1 serves as the primary measurement framework. To enable survey-based measurement, each maturity level is operationalized through a set of observable, measurable indicators, as presented in Table 2.1.

Respondents were asked to select the description that best matched their organization's current PMO IS integration approach. The chosen level functions as the primary independent variable in the study.

2.3. Survey Instrument Design

The survey instrument was designed to measure the core constructs of the study: integration maturity level, reporting automation, data quality, and governance standardization. The instrument consists of four sections.

Table 2.1. Operationalization of Integration Maturity Model levels

Level	Name	Observable Indicators	Survey Items
1	Initial	No system connections; all data consolidated manually in spreadsheets; no automated reports	“Data is consolidated manually from multiple tools into spreadsheets”
2	Developing	Some file-based exports between systems; partial automation; scheduled report generation	“Some systems share data via file exports (CSV/Excel)”
3	Defined	API connections between key systems; automated data flows; near real-time reporting	“Systems are connected via APIs; reports are generated automatically”
4	Optimized	Full platform integration; unified BI dashboard; real-time portfolio visibility	“All systems are fully integrated in a unified platform with real-time dashboards”

- **Part A — Organizational Profile:** Questions about the organization’s size, industry, and PMO structure (categorical items).
- **Part B — Integration Maturity Assessment:** A single categorical item in which respondents select the IMM level that best describes their PMO’s IS integration approach, supplemented by Likert-scale items measuring specific integration practices.
- **Part C — Reporting Automation and Data Quality:** Five-point Likert-scale items measuring the degree of reporting automation and perceived data accuracy and timeliness.
- **Part D — Governance Standardization:** Five-point Likert-scale items measuring the degree of governance standardization in PMO processes.

All Likert-scale items use a five-point response scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The survey was designed in English and distributed electronically via Google Forms. The estimated completion time was 10–15 minutes.

2.4. Survey Questionnaire

The complete survey instrument is presented in Table 2.2. Each item is aligned with a specific research construct and hypothesis.

Note on scaling. All items in Parts B–D use a five-point Likert scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree. Item C5 is reverse scored, meaning higher values indicate lower data quality issues.

2.5. Expert Interview Design

To supplement the quantitative survey findings with qualitative depth, one structured expert interview was conducted with an experienced project management professional from a Latvian enterprise. The interview followed a structured protocol with open-ended questions covering the following themes:

- The current state of IS integration in the respondent's PMO environment
- Challenges experienced due to low integration maturity
- The role of BI and dashboard systems in governance standardization
- Perceived barriers to advancing integration maturity
- Practical recommendations for improving PMO IS integration

Interview responses were analyzed qualitatively using thematic analysis to identify patterns and insights that complement the quantitative findings. In accordance with the ethical principles outlined in Section 2.9, the interview participant is referred to anonymously in the thesis.

2.6. Data Collection Procedure

The survey was distributed to representatives of medium and large Latvian enterprises (50+ employees) with formal PMO structures. The target sample size was 30–50 responses, which is sufficient for the planned descriptive and correlational statistical analyses [4]. Respondents were identified through professional networks, LinkedIn, and direct organizational contacts. Only one response per organization was accepted to ensure independence of observations.

Data collection took place over a period of four to six weeks. All responses are anonymous and stored securely in accordance with data protection principles. Participation was voluntary and respondents were informed of the purpose of the study prior to completing the survey.

2.7. Data Analysis Methods

The collected data was analyzed using the following statistical methods:

Descriptive statistics. Frequency distributions and percentages were used to describe the distribution of integration maturity levels across the sample. Mean scores and standard deviations were calculated for Likert-scale items measuring reporting automation, data quality, and governance standardization.

Normality testing. The Shapiro-Wilk test was applied to determine whether the data follows a normal distribution, which informed the choice of correlation method [4]. As the data did not follow a normal distribution, Spearman rank correlation was selected as the appropriate non-parametric method.

Correlation analysis. Spearman rank correlation coefficients were calculated to examine the relationships between integration maturity level and the three dependent variables: reporting automation, data quality, and governance standardization. Statistical significance was assessed at the $p < 0.05$ level.

Qualitative analysis. The expert interview transcript was analyzed using thematic analysis to identify recurring themes and insights that supplement the quantitative findings.

All quantitative analyses were conducted using SPSS statistical software.

2.8. Validity and Reliability

To ensure the quality of the research instrument and findings, the following measures were taken to address validity and reliability.

Content validity. The survey items were designed based on established theoretical frameworks — specifically the Integration Maturity Model proposed in Chapter 1, the CMMI framework [10], and the data quality dimensions from DAMA-DMBOK [2]. Each survey item is explicitly linked to a research construct and hypothesis, ensuring that the instrument measures what it is intended to measure.

Construct validity. The survey constructs — integration maturity, reporting automation, data quality, and governance standardization — are operationalized based on definitions from peer-reviewed literature and established frameworks. Multiple items per construct were included to reduce measurement error and improve construct coverage.

Internal consistency reliability. Cronbach's alpha was calculated for each multi-item construct scale to assess internal consistency. All constructs returned values above the

accepted threshold of 0.70 [4], confirming satisfactory internal consistency of the survey instrument.

External validity. The study targets a specific population — representatives of medium and large Latvian enterprises with formal PMO structures. While the findings may not be directly generalizable to all organizations globally, they are intended to be representative of the Latvian enterprise context within the defined scope.

2.9. Research Ethics

This study adheres to the following ethical principles throughout the data collection and analysis process:

- **Voluntary participation.** All survey respondents and the interview participant participated voluntarily and were free to withdraw at any time without consequence.
- **Informed consent.** Respondents were informed of the purpose of the study, how their data would be used, and who would have access to the results before completing the survey.
- **Anonymity and confidentiality.** Survey responses were collected anonymously. No personally identifiable information was collected or stored. The interview participant is anonymized in the thesis.
- **Data protection.** All collected data is stored securely and processed in accordance with the General Data Protection Regulation (GDPR) and applicable Latvian data protection legislation.
- **Honest reporting.** All findings are reported accurately and without manipulation, including results that do not support the research hypotheses.

2.10. Limitations of the Research Design

While this study is designed to provide a rigorous empirical assessment of IS integration maturity in PMO environments in Latvia, several limitations must be acknowledged.

Self-reported data. The survey relies on self-reported perceptions of PMO representatives, which may be subject to social desirability bias — respondents may rate their organization's integration maturity higher than it actually is. To mitigate this, the survey includes specific behavioral indicators rather than general self-assessment questions.

Sample size. The target sample of 30–50 responses is sufficient for descriptive and correlational analysis but limits the statistical power of more advanced models and prevents subgroup analysis by industry or organization size. Future research with larger samples would strengthen the generalizability of findings.

Cross-sectional design. The study captures a snapshot of integration maturity at a single point in time and cannot establish causal relationships between integration maturity and the dependent variables. Longitudinal research would be needed to confirm causal direction.

Geographic scope. The study is deliberately bounded to Latvian enterprises, which limits the generalizability of findings to other national or regional contexts. However, this scope is appropriate given the research aim of providing Latvia-specific empirical evidence.

Construct measurement. Integration maturity is measured through a self-developed instrument rather than a previously validated scale, as no existing validated scale specifically measures IS integration maturity in PMO environments. While the instrument is grounded in established theoretical frameworks, its psychometric properties require further validation in future research.

2.11. Chapter Summary

This chapter described the research methodology applied in this thesis. A quantitative cross-sectional survey design complemented by one qualitative expert interview was adopted. The four-level Integration Maturity Model was operationalized into measurable survey indicators. The survey instrument was designed to measure integration maturity, reporting automation, data quality, and governance standardization across PMO environments in Latvian enterprises. A total of 40 valid responses were collected from medium and large Latvian enterprises with formal PMO structures. The Shapiro-Wilk normality test indicated non-normal data distribution, leading to the use of Spearman rank correlation analysis to test the four research hypotheses. Validity, reliability, ethical principles, and research limitations were also addressed. The results of the empirical study are presented in Chapter 3.

Table 2.2. Survey instrument items by construct and hypothesis

No.	Survey Item	Construct	Hypothesis
Part A — Organizational Profile			
A1	How many employees does your organization have? (50–249 / 250+)	Organization size	–
A2	What industry does your organization operate in? (categorical)	Industry sector	–
A3	How long has your PMO been formally established? (years)	PMO maturity	–
A4	How many active projects does your PMO currently oversee? (numerical)	PMO scope	–
Part B — Integration Maturity Assessment			
B1	Which of the following best describes how data flows between your PMO’s information systems? (select one IMM level description)	Integration maturity level	H1
B2	Our PMO systems share data automatically without manual intervention.	Integration automation	H1, H2
B3	Our PMO uses API connections between project tracking, financial, and reporting tools.	API integration	H2
B4	Our PMO relies on manual data entry or file exports (CSV/Excel) to consolidate project data.	Manual integration	H1, H2
B5	Our PMO operates within a unified platform ecosystem where all systems are fully integrated.	Platform integration	H1, H2
Part C — Reporting Automation and Data Quality			
C1	Project status reports in our PMO are generated automatically from integrated systems.	Reporting automation	H2
C2	Our PMO dashboards are updated in real time without manual intervention.	Reporting automation	H2
C3	The data in our PMO reports is accurate and reliable.	Data accuracy	H3
C4	Project data is available in a timely manner for decision-making.	Data timeliness	H3
C5	We frequently encounter errors or inconsistencies in our PMO reporting data. (reverse scored)	Data quality	H3
Part D — Governance Standardization			
D1	Our PMO follows standardized processes for project reporting across all projects.	Governance standardization	H4
D2	Our PMO uses centralized dashboards to monitor all projects in one place.	BI centralization	H4
D3	Project governance processes in our PMO are consistently applied across	Governance consistency	H4

3. Empirical Study

This chapter presents the findings of the empirical part of the study. It includes the survey data gathered from forty medium-sized and large Latvian enterprises with formal PMO structures, along with findings from one structured expert interview. The chapter is organized into five sections: sample characteristics, descriptive statistics, hypothesis testing, expert interview findings, and a chapter summary.

3.1. Sample Characteristics

A total of forty valid survey responses were collected, satisfying the 30–50 response target specified in the research methodology. The respondents represented organizations from a variety of industries and sizes, providing a diverse and representative sample for the analysis.

3.1.1. Organization Size

Over half of the respondents were from medium-sized enterprises. Specifically, 52.5% (n=21) reported having 50–249 employees, 35% (n=14) had 250–999 employees, and 12.5% (n=5) represented large organizations with 1000 or more employees. This distribution reflects the actual business landscape in Latvia, which is characterized by a dominance of medium-sized companies.

3.1.2. Industry Distribution

The sample covered seven different industry sectors. The largest group was Information Technology at 37.5% (n=15), which is not surprising given that IT companies tend to have more formal PMO structures. Finance and Banking, Retail and Commerce, and Telecommunications each accounted for 12.5% (n=5). Public Sector and Other industries each represented 10% (n=4), while Manufacturing made up 5% (n=2).

3.1.3. PMO Maturity and Scope

Respondents represented organizations with varying levels of PMO establishment and project portfolio sizes. This diversity strengthens the reliability of the findings by ensuring that the results are not skewed toward any single type of organization.

3.2. Descriptive Statistics

3.2.1. Integration Maturity Level Distribution

One of the central questions of this study was to determine at which integration maturity level most Latvian PMO environments currently operate. The results are presented in Table 3.1.

Table 3.1. Distribution of IS Integration Maturity Levels

Level	Name	Frequency	Percentage
1	Initial	6	15.0%
2	Developing	13	32.5%
3	Defined	15	37.5%
4	Optimized	6	15.0%
Total		40	100%

The results clearly show that the majority of PMO environments (70%, n=28) operate at intermediate maturity levels — either Level 2 (Developing) or Level 3 (Defined). Only 15% (n=6) have reached the highest level of integration maturity, and another 15% (n=6) remain at the most basic level. This pattern suggests that while many Latvian enterprises have moved beyond entirely manual integration, full platform-level integration remains uncommon.

3.2.2. Descriptive Statistics for Key Constructs

Table 3.2 shows the mean scores and standard deviations for the three dependent variables — reporting automation, data quality, and governance standardization — measured using five-point Likert scales.

Table 3.2. Descriptive Statistics for Key Constructs

Construct	N	Mean	Std Dev	Min	Max
Reporting Automation	40	3.38	0.85	1.00	5.00
Data Quality	40	3.32	0.73	2.00	5.00
Governance Standardization	40	3.64	0.71	1.00	5.00

All three constructs show moderate mean scores, suggesting that the sampled organizations have achieved a moderate but not yet high level of performance in these areas. Governance standardization recorded the highest mean (M=3.64), while data quality had the

lowest ($M=3.32$). The standard deviations indicate a reasonable spread of responses, reflecting genuine variation in IS integration practices across the sample.

3.3. Hypothesis Testing

Before conducting correlation analysis, the Shapiro-Wilk normality test was applied to all variables. The results indicated that the data did not follow a normal distribution, which led to the use of Spearman rank correlation — a non-parametric method suitable for ordinal and non-normally distributed data [4]. Statistical significance was assessed at the $p < 0.05$ level.

3.3.1. Hypothesis 1

H1: The majority of PMOs in medium and large Latvian enterprises operate at intermediate integration maturity levels (Levels 2 or 3) rather than at the most advanced level.

As shown in Table 3.1, 70% of respondents ($n=28$) reported operating at Level 2 or Level 3, while only 15% ($n=6$) had reached Level 4. This clearly confirms that most Latvian PMO environments are at intermediate rather than advanced integration maturity levels. **Hypothesis 1 is supported.**

3.3.2. Hypothesis 2

H2: PMOs that use API or platform-based integration report higher levels of reporting automation compared to those that rely on manual data handling.

Spearman correlation analysis showed a statistically significant positive relationship between integration maturity level and reporting automation ($r=0.405$, $p=0.010$). This means that organizations with higher IS integration maturity consistently reported higher levels of automated reporting. **Hypothesis 2 is supported.**

3.3.3. Hypothesis 3

H3: Higher integration maturity is associated with better data accuracy and faster access to project information.

A statistically significant positive correlation was found between integration maturity level and data quality ($r=0.321$, $p=0.043$). Although this is the weakest of the three correlations, it remains statistically significant and confirms that better integrated systems are associated with higher perceived data accuracy and timeliness. **Hypothesis 3 is supported.**

3.3.4. Hypothesis 4

H4: PMOs that use centralized BI or dashboard tools tend to have more standardized governance practices.

The correlation analysis revealed a statistically significant positive relationship between integration maturity level and governance standardization ($r=0.403$, $p=0.010$). Organizations with higher integration maturity demonstrated notably stronger governance standardization practices. **Hypothesis 4 is supported.**

Table 3.3. Summary of Hypothesis Testing Results

H	Hypothesis	r	p	Result
H1	Most PMOs at Levels 2–3	–	–	Supported
H2	Maturity → Reporting Automation	0.405	0.010	Supported
H3	Maturity → Data Quality	0.321	0.043	Supported
H4	Maturity → Governance	0.403	0.010	Supported

3.4. Expert Interview Findings

One structured expert interview was conducted to add qualitative depth to the survey findings. The participant is a professional with experience in PMO environments. In line with the ethical principles outlined in Chapter 2, the participant is referred to anonymously as Expert 1.

3.4.1. Expert 1 — Project Management Professional

Expert 1 is a skilled project management professional working in a PMO environment. Their organization operates a hybrid-integrated IS ecosystem, utilizing tools such as ServiceNow and/or Planview for strategic portfolio management, Jira for agile execution, and Microsoft Project Online for waterfall-style schedules. These systems are connected through APIs and middleware solutions, placing the organization at a high integration maturity level (Level 3–4 of the IMM).

Despite this high level of integration, Expert 1 identified three recurring challenges. First, data latency — although systems are connected, short delays between execution tools and the portfolio layer can cause stakeholders to view slightly outdated reports. Second, semantic misalignment — different departments define concepts such as progress in different ways, requiring ongoing reconciliation between technical and financial systems. Third,

legacy resistance — older ERP systems used by the Finance department are difficult to connect with modern cloud-based tools, sometimes requiring manual data exports as a workaround.

Regarding reporting, Expert 1 described a 70/30 split in favour of automation. Standard KPIs, budget consumption, and resource utilization are pulled automatically into Power BI and Tableau dashboards, while the remaining 30% — primarily executive summaries and qualitative narratives — still requires human input.

On data quality, Expert 1 rated satisfaction as 4 out of 5. Timeliness had improved significantly due to automated syncing, but accuracy remains dependent on how consistently project managers update their data. To address this, the organization introduced a Data Health Score system to monitor data hygiene across projects.

Expert 1 confirmed that a centralized Power BI Command Center had been a major step forward for governance standardization. It enforced universal KPI definitions across all projects, eliminated subjective reporting, and made project performance visible to all executives at the same time.

For recommendations, Expert 1 suggested that Latvian enterprises should consider modular, low-code platforms such as Monday.com or ClickUp, which offer strong integration capabilities at a lower cost than enterprise-level tools. Expert 1 also highlighted the need for standardized connectors between local Latvian accounting software and global PM tools, and stressed the importance of data literacy training for Latvian project leaders.

3.5. Chapter Summary

This chapter presented the empirical findings of the study. Based on 40 valid survey responses from medium and large Latvian enterprises, the results showed that 70% of PMO environments operate at intermediate integration maturity levels (Levels 2 and 3), which confirms Hypothesis 1. Statistically significant positive correlations were found between integration maturity and all three dependent variables: reporting automation ($r=0.405$, $p=0.010$), data quality ($r=0.321$, $p=0.043$), and governance standardization ($r=0.403$, $p=0.010$), confirming Hypotheses 2, 3, and 4. The expert interview with Expert 1 provided rich qualitative support for these findings, particularly highlighting the value of centralized BI systems for governance standardization and the ongoing challenge of legacy system integration. The overall findings are discussed and interpreted in Chapter 4.

4. Discussion and Recommendations

This chapter interprets the empirical findings presented in Chapter 3 in the context of the theoretical framework and existing literature reviewed in Chapter 1. It examines the significance of the results, compares them with prior research, proposes practical recommendations for improving information system integration maturity in PMO environments in Latvian enterprises, and reflects on unexpected findings and theoretical implications.

4.1. Discussion of Findings

4.1.1. Integration Maturity Distribution in Latvian PMOs

According to the survey findings, 70% of the sampled PMO environments operate at intermediate integration maturity levels — either Level 2 (Developing) or Level 3 (Defined). This result supports Hypothesis 1 and is consistent with broader research on digital maturity in Baltic enterprises, which indicates that Latvian organizations generally fall behind their Nordic counterparts in terms of digital integration and automation [3].

The prevalence of Levels 2 and 3 indicates that most Latvian enterprises have progressed beyond entirely manual data consolidation, yet full platform-level integration has not been achieved. This is a crucial observation, as it reveals that most organizations are in a transitional phase — they have acknowledged the need for integration but have not yet committed fully to reaching Level 4. This result also aligns with the expert interview, in which Expert 1 noted that challenges such as data latency and legacy system resistance persist even in organizations with relatively high integration maturity.

4.1.2. Integration Maturity and Reporting Automation

The statistically significant positive correlation between integration maturity and reporting automation ($r=0.405$, $p=0.010$) confirms that organizations with more advanced IS integration produce reports more automatically. This supports Hypothesis 2 and is consistent with research by Hohpe and Woolf [6], who demonstrated that API-based integration architectures substantially reduce the need for manual data handling.

The practical implication is straightforward: organizations that invest in higher IS integration levels can expect a notable reduction in the time and effort devoted to manual reporting. Expert 1 confirmed this, describing a 70/30 split in favour of automated reporting

in their organization, compared to what is likely a considerably lower automation rate in Level 1 and Level 2 organizations.

4.1.3. Integration Maturity and Data Quality

The positive correlation between integration maturity and data quality ($r=0.321$, $p=0.043$), while the weakest of the three, remains statistically significant. This confirms Hypothesis 3 and supports the theoretical argument presented in Chapter 1 that manual data consolidation introduces errors and delays that compromise data quality [2].

The comparatively weaker correlation for data quality may reflect the fact that it is shaped by many factors beyond IS integration alone — including user behaviour, data entry discipline, and organizational culture. Expert 1 emphasized this point, noting that even at high integration maturity, data quality depends heavily on how consistently project managers update their records. This suggests that technical integration must be accompanied by organizational practices that actively support data hygiene.

4.1.4. Integration Maturity and Governance Standardization

The strong positive correlation between integration maturity and governance standardization ($r=0.403$, $p=0.010$) confirms Hypothesis 4 and provides empirical support for the theoretical link between centralized BI systems and governance quality established in Chapter 1 [9]. Organizations with higher integration maturity demonstrated noticeably stronger governance standardization practices.

The expert interview strongly supports this finding. Expert 1 described how the introduction of a centralized Power BI Command Center transformed governance in their organization by enforcing uniform KPI definitions, eliminating subjective reporting, and making project performance transparently visible to all executives at the same time. This qualitative evidence helps explain the mechanism through which higher integration maturity leads to improved governance standardization.

4.2. Comparison with Existing Literature

The findings of this study are broadly consistent with existing research on IS integration and PMO maturity. The dominance of intermediate maturity levels in the sample aligns with research by Hobbs and Aubry [5], who found that most PMOs struggle to advance beyond basic standardization practices. The positive relationships between integration maturity and the three dependent variables are consistent with the theoretical predictions of

the Integration Maturity Model proposed in Chapter 1.

Nevertheless, this study makes a distinctive contribution by providing empirical evidence specifically from the Latvian enterprise context. As noted in Chapter 1, no prior study had examined IS integration maturity in PMO environments within Latvia. The findings suggest that Latvian enterprises follow a pattern similar to those documented in broader European research, but with a tendency toward lower overall maturity levels, which is consistent with the digital maturity gap between Baltic and Nordic countries identified by Eurostat [3].

4.3. Unexpected Findings and Theoretical Implications

4.3.1. Unexpected Findings

A notable finding was that data quality showed the weakest correlation with integration maturity ($r=0.321$) compared to reporting automation ($r=0.405$) and governance standardization ($r=0.403$). This was somewhat unexpected, as the theoretical framework suggested that automated integration would directly improve data accuracy by eliminating manual entry errors. However, as Expert 1 observed, data quality is also strongly shaped by human behaviour and organizational discipline — factors that operate independently of technical integration level. This suggests that the relationship between IS integration and data quality is more complex than initially anticipated and warrants further investigation.

Another interesting observation is that 15% of organizations operated at Level 4 — the highest maturity level. This proportion is higher than might be expected given the generally lower digital maturity of Baltic enterprises relative to Nordic counterparts. This may reflect a degree of self-selection bias, as organizations with more advanced PMO structures may have been more willing to participate in a PMO-focused survey.

4.3.2. Theoretical Implications

This study advances the theoretical understanding of IS integration maturity in PMO environments in several ways. First, it provides empirical validation of the four-level Integration Maturity Model proposed in Chapter 1, demonstrating that its levels correspond to meaningful differences in reporting automation, data quality, and governance standardization. Second, it extends the existing body of literature on digital maturity models by applying the maturity concept specifically to PMO IS integration — a context that had not previously been examined empirically. Third, it contributes the first empirical evidence of IS integration maturity patterns in Latvian enterprises, adding to the growing body of research on digital

transformation in Baltic and Eastern European contexts.

4.4. Practical Recommendations

Based on the empirical findings and expert interview insights, the following practical recommendations are proposed for medium and large Latvian enterprises seeking to improve their PMO IS integration maturity.

4.4.1. Recommendation 1 — Prioritize API-Based Integration

Organizations currently at Level 1 or Level 2 should focus on establishing API-based connections between their core PMO systems — particularly between project tracking tools, financial systems, and reporting platforms. Even basic API integration can substantially reduce manual data consolidation and improve reporting automation, as supported by the strong correlation between maturity level and reporting automation found in this study.

4.4.2. Recommendation 2 — Invest in Centralized BI Dashboards

One of the most effective steps organizations can take to strengthen governance standardization is to adopt a centralized BI or dashboard solution. Tools such as Power BI, Tableau, or Smartsheet can bring together data from multiple systems into a unified view, enforcing consistent KPI definitions and improving transparency across the organization. As Expert 1 noted, this single change can be transformative for governance quality.

4.4.3. Recommendation 3 — Consider Low-Code Integration Platforms

For medium-sized Latvian enterprises that lack the resources for large-scale enterprise integration, low-code and modular platforms such as Monday.com, ClickUp, or Zapier offer a cost-effective route toward higher integration maturity. These tools enable organizations to connect existing systems without requiring extensive technical expertise or significant financial investment.

4.4.4. Recommendation 4 — Address Local System Interoperability

Many Latvian enterprises rely on local accounting and ERP software that is not natively compatible with global PM tools. Organizations should prioritize developing or procuring standardized connectors between these local systems and their PMO tools to eliminate the manual data exports that currently fill this gap.

4.4.5. Recommendation 5 — Invest in Data Literacy

Technical integration alone is insufficient to improve data quality. Organizations must also invest in training project managers and PMO staff to appreciate the importance of data hygiene and to use integrated systems effectively. As Expert 1 observed, the introduction of a Data Health Score system helped sustain data accuracy even in a highly integrated environment.

4.5. Limitations of the Study

Although the findings of this study are meaningful and statistically significant, several limitations should be kept in mind when interpreting the results. The sample of 40 respondents is adequate for the planned analyses but restricts the statistical power of more advanced models. Reliance on self-reported data means that respondents may have overestimated their organization's integration maturity. The cross-sectional design captures only a single point in time and cannot establish causal relationships between variables. Finally, the dominance of IT sector organizations in the sample may limit the generalizability of findings to other industries. These limitations are discussed in greater detail in Chapter 2.

4.6. Chapter Summary

This chapter examined the empirical findings of the study within the context of the theoretical framework and existing literature. All four hypotheses were confirmed by the data, with statistically significant positive correlations identified between IS integration maturity and reporting automation, data quality, and governance standardization. The findings align with broader research on PMO maturity and digital transformation in Baltic enterprises. A noteworthy result was the comparatively weak correlation between integration maturity and data quality, suggesting that human behaviour and organizational culture play an important role alongside technical integration. Five practical recommendations were developed to support Latvian enterprises in advancing their PMO IS integration maturity. These findings and recommendations provide the foundation for the conclusions presented in the following chapter.

5. Future Work and Broader Reflections

This chapter offers broader reflections on the findings of this study and suggests directions for future research. While the core findings and recommendations were presented in Chapters 3 and 4, this chapter looks beyond the immediate scope of the thesis to consider wider implications and open questions.

5.1. Future Research Directions

This study provides a first empirical assessment of IS integration maturity in PMO environments in Latvia. However, several important research directions remain unexplored and could be pursued in future work.

Longitudinal research. The current study captures a snapshot of integration maturity at a single point in time. A longitudinal study tracking the same organizations over several years would provide valuable insights into how integration maturity evolves and what factors drive or hinder progression between maturity levels.

Broader Baltic comparison. Extending this study to include Estonia and Lithuania would allow for a meaningful comparison of IS integration maturity patterns across the Baltic region. Given that Estonia is widely regarded as one of the most digitally advanced countries in Europe, such a comparison could reveal important differences and lessons for Latvian enterprises.

Industry-specific benchmarks. The current study treats all industries as comparable. Future research could develop industry-specific IS integration maturity benchmarks for Latvia, recognizing that the integration needs and challenges of IT companies differ significantly from those of manufacturing or public sector organizations.

SME-focused research. This study targeted enterprises with 50 or more employees. However, many Latvian enterprises are smaller than this threshold. Future research could examine IS integration maturity challenges specific to small and micro enterprises, which face different resource constraints and may require tailored integration solutions.

Role of emerging technologies. The rapid development of artificial intelligence, machine learning, and low-code platforms is likely to significantly reshape IS integration in PMO environments in the coming years. Future research could examine how these technologies affect integration maturity progression and what new maturity frameworks may be

needed to capture their impact.

5.2. Broader Societal and Economic Implications

The findings of this study have implications that extend beyond individual organizations. Latvia's digital transformation agenda, supported by EU structural funds and national digitalization strategies, places increasing emphasis on improving the digital capabilities of Latvian enterprises. The finding that 70% of PMO environments operate at intermediate integration maturity levels suggests that significant potential remains untapped.

Improving IS integration maturity in Latvian PMO environments could generate wider economic benefits, including greater project delivery efficiency, more effective use of public and private investment, and stronger alignment between organizational strategy and project execution. As Latvia continues to develop its digital economy, investments in PMO IS integration may play an important role in strengthening organizational competitiveness within the European Union.

Furthermore, the recommendations proposed in Chapter 4 — particularly the emphasis on low-code platforms and data literacy training — align well with the EU's broader digital skills agenda and could be incorporated into national programmes supporting the digital transformation of Latvian enterprises.

5.3. Chapter Summary

This chapter presented future research directions and broader reflections on the implications of the study. Key directions for future research include longitudinal studies, Baltic regional comparisons, industry-specific benchmarking, SME-focused research, and investigation of the role of emerging technologies in PMO IS integration. The findings of this study also carry wider societal and economic implications for Latvia's digital transformation agenda, suggesting that improving PMO IS integration maturity could contribute meaningfully to national competitiveness and organizational effectiveness.

Conclusions

The central aim of this thesis was to evaluate how well information systems are integrated within Project Management Office environments in medium and large Latvian enterprises, and to explore whether this integration maturity has any meaningful connection to reporting automation, data quality, and governance standardization. This concluding section confirms the completion of all research tasks, reflects on the four hypotheses, and assesses whether the overall aim of the thesis has been achieved.

Confirmation of research tasks.

Task 1 was addressed in Chapter 1, where a thorough review of existing literature was carried out covering PMO digitalization, information system integration architectures, and maturity models applicable to IS environments. This review laid the conceptual and theoretical groundwork for the entire study.

Task 2 was also completed in Chapter 1, where a four-level Integration Maturity Model was constructed and made measurable through clearly defined criteria at each level, encompassing integration mechanisms, degree of automation, reporting structure, and the extent of BI centralization.

Task 3 was fulfilled in Chapter 2, which documented the development and validation of a structured survey instrument. The instrument included both Likert-scale and categorical items, each directly tied to the dimensions of the Integration Maturity Model.

Task 4 involved distributing the survey to representatives of medium and large Latvian enterprises employing 50 or more people and maintaining a formal PMO structure. In total, 40 usable responses were received, which fell within the 30–50 target range established in the methodology.

Task 5 was carried out in Chapter 3, where a structured expert interview was held with a seasoned project management professional. The interview served to enrich the quantitative survey findings with qualitative perspectives and practical architectural insights.

Task 6 was also completed in Chapter 3, where descriptive statistical analysis was used to map out how integration maturity levels were distributed across the organizations in the sample.

Task 7 was fulfilled in Chapter 3 through the application of Spearman rank correla-

tion analysis, which were used to investigate the relationships between integration maturity and the three dependent variables: reporting automation, data accuracy, and governance standardization.

Task 8 was addressed in Chapter 4, where the empirical results were interpreted, conclusions were drawn in relation to all four hypotheses, and five concrete recommendations were put forward to help Latvian enterprises strengthen their PMO IS integration maturity.

Validation of hypotheses.

All four hypotheses received support from the empirical data gathered in this study.

Hypothesis 1 was confirmed. The survey data showed that 70% of the sampled PMO environments are operating at intermediate integration maturity levels — specifically Level 2 (Developing) or Level 3 (Defined) — rather than at the most advanced stage. With only 15% of organizations having reached Level 4, it is clear that the majority of Latvian PMOs remain in a transitional phase of IS integration development.

Hypothesis 2 was confirmed. A statistically significant positive correlation was identified between integration maturity and reporting automation ($r=0.405$, $p=0.010$). This finding indicates that organizations operating at higher integration maturity levels consistently achieve greater degrees of automated reporting, thereby reducing their dependence on manual data consolidation.

Hypothesis 3 was confirmed. A statistically significant positive correlation was found between integration maturity and data quality ($r=0.321$, $p=0.043$). Although this was the weakest association among the three, it nonetheless confirms that organizations with better integrated systems tend to perceive their data as more accurate and accessible in a timely manner.

Hypothesis 4 was confirmed. A statistically significant positive correlation was established between integration maturity and governance standardization ($r=0.403$, $p=0.010$). Organizations with higher integration maturity exhibited noticeably stronger governance standardization, a finding further reinforced by the expert interview, which highlighted the transformative role of centralized BI dashboards in bringing consistency to governance processes.

Achievement of the aim.

The aim of this thesis has been fully achieved. Through a combination of quantitative survey analysis and qualitative expert interview findings, the study successfully assessed IS integration maturity across PMO environments in medium and large Latvian enterprises. The

results clearly demonstrate statistically significant relationships between integration maturity and all three dependent variables. It can be concluded that most Latvian PMO environments currently operate at intermediate maturity levels, and that organizations with higher integration maturity consistently perform better in terms of reporting automation, data quality, and governance standardization. The aim of the thesis has therefore been reached.

Practical and theoretical value.

This study offers value on both a practical and theoretical level. From a practical standpoint, five evidence-based recommendations were developed to guide Latvian enterprises in improving their PMO IS integration maturity. These cover a range of actionable steps: building API-based integration between core systems, adopting centralized BI dashboards, exploring low-code platforms as a cost-effective alternative for medium-sized organizations, resolving compatibility issues between local Latvian accounting software and international PM tools, and investing in data literacy programmes for project management staff.

From a theoretical standpoint, this study delivers the first empirical validation of a PMO-specific IS Integration Maturity Model within the Latvian enterprise context. It demonstrates that maturity modeling can be meaningfully applied to IS integration in PMO settings, and adds a new layer of empirical evidence to the growing body of scholarship on digital transformation in Baltic and Eastern European organizations.

All eight tasks outlined in the Introduction were successfully completed. Tasks 1 and 2 were addressed in Chapter 1, Task 3 in Chapter 2, Tasks 4–7 in Chapter 3, and Task 8 in Chapter 4. All four research hypotheses were validated by the empirical data, and the aim of the thesis has been fully achieved.

Literature

- [1] “Commission recommendation of 6 may 2003 concerning the definition of micro, small and medium-sized enterprises,” European Commission, Accessed: Mar. 22, 2026. [Online]. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32003H0361>
- [2] DAMA International, *DAMA-DMBOK: Data Management Body of Knowledge*, 2nd ed. Basking Ridge, NJ: Technics Publications, 2017.
- [3] “Digital economy and society statistics – enterprises,” Eurostat, Accessed: Mar. 22, 2026. [Online]. Available: https://ec.europa.eu/eurostat/statistics-explained/index.php/Digital_economy_and_society_statistics_-_enterprises
- [4] A. Field, *Discovering Statistics Using IBM SPSS Statistics*, 5th ed. London: SAGE Publications, 2018.
- [5] B. Hobbs and M. Aubry, “A multi-phase research program investigating project management offices: The results of phase 1,” *Project Management Journal*, vol. 38, no. 1, pp. 74–86, 2007.
- [6] G. Hohpe and B. Woolf, *Enterprise Integration Patterns: Designing, Building, and Deploying Messaging Solutions*. Boston: Addison-Wesley, 2004.
- [7] K. C. Laudon and J. P. Laudon, *Management Information Systems: Managing the Digital Firm*, 14th ed. Boston: Pearson, 2015.
- [8] “Magic quadrant for analytics and business intelligence platforms,” Gartner, Accessed: Mar. 22, 2026. [Online]. Available: <https://www.gartner.com/en/documents/3982949>
- [9] S. Negash, “Business intelligence,” *Communications of the Association for Information Systems*, vol. 13, pp. 177–195, 2004.
- [10] M. C. Paulk, B. Curtis, M. B. Chrissis, and C. V. Weber, *Capability Maturity Model for Software, Version 1.1*. Pittsburgh, PA: Software Engineering Institute, Carnegie Mellon University, 1993.

- [11] Project Management Institute, *A Guide to the Project Management Body of Knowledge (PMBOK Guide)*, 7th ed. Newtown Square, PA: Project Management Institute, 2021.
- [12] M. Saunders, P. Lewis, and A. Thornhill, *Research Methods for Business Students*, 8th ed. Harlow: Pearson, 2019.

A. Survey Questionnaire

This appendix presents the complete survey instrument used to collect empirical data for this study. The survey was distributed electronically via Google Forms to representatives of medium and large Latvian enterprises with formal Project Management Office (PMO) structures. Participation was voluntary and all responses were collected anonymously.

Survey title: Assessment of Information System Integration Maturity in Project Management Offices

Estimated completion time: 10–15 minutes

Target respondents: PMO managers, project managers, IT managers, or digital transformation leads at organizations with 50 or more employees and a formal PMO structure.

Part A — Organizational Profile

A1. How many employees does your organization currently have?

- ☐ 50–249 employees (medium-sized enterprise)
- ☐ 250–999 employees (large enterprise)
- ☐ 1000 or more employees (very large enterprise)

A2. What industry does your organization primarily operate in?

- ☐ Information Technology
- ☐ Finance and Banking
- ☐ Manufacturing
- ☐ Retail and Commerce
- ☐ Telecommunications
- ☐ Public Sector
- ☐ Other (please specify): _____

A3. How long has your PMO been formally established?

- ☐ Less than 1 year
- ☐ 1–3 years
- ☐ 3–5 years
- ☐ More than 5 years

A4. How many active projects does your PMO currently oversee?

- ☐ 1–10 projects
- ☐ 11–25 projects
- ☐ 26–50 projects
- ☐ More than 50 projects

Part B — Integration Maturity Assessment

B1. Which of the following best describes how data flows between your PMO’s information systems? *(Select one option.)*

- ☐ **Level 1 — Initial:** Data is consolidated manually from multiple tools into spreadsheets. There are no automated connections between systems.
- ☐ **Level 2 — Developing:** Some systems share data via scheduled file exports (e.g., CSV or Excel). Some manual work is still required to consolidate and verify data.
- ☐ **Level 3 — Defined:** Systems are connected via APIs. Reports are generated automatically and data flows are mostly automated and near real-time.
- ☐ **Level 4 — Optimized:** All systems are fully integrated within a unified platform with real-time dashboards and full portfolio visibility. Governance processes are fully standardized.

For items B2–B5, please indicate your level of agreement using the following scale:

1 2 3 4 5
 Strongly Disagree Disagree Neutral Agree Strongly Agree

No.	Statement	Rating (1–5)
B2	Our PMO systems share data automatically without manual intervention.	_____
B3	Our PMO uses API connections between project tracking, financial, and reporting tools.	_____
B4	Our PMO relies on manual data entry or file exports (CSV/Excel) to consolidate project data.	_____
B5	Our PMO operates within a unified platform ecosystem where all systems are fully integrated.	_____

Part C — Reporting Automation and Data Quality

Please indicate your level of agreement with each statement (1 = Strongly Disagree, 5 = Strongly Agree). Item C5 is reverse scored.

No.	Statement	Rating (1–5)
C1	Project status reports in our PMO are generated automatically from integrated systems.	_____
C2	Our PMO dashboards are updated in real time without manual intervention.	_____
C3	The data in our PMO reports is accurate and reliable.	_____
C4	Project data is available in a timely manner for decision-making.	_____
C5	We frequently encounter errors or inconsistencies in our PMO reporting data. <i>(reverse scored)</i>	_____

Part D — Governance Standardization

Please indicate your level of agreement with each statement (1 = Strongly Disagree, 5 = Strongly Agree).

No.	Statement	Rating (1–5)
D1	Our PMO follows standardized processes for project reporting across all projects.	_____
D2	Our PMO uses centralized dashboards to monitor all projects in one place.	_____
D3	Project governance processes in our PMO are consistently applied across all departments.	_____
D4	Our PMO has clearly defined and documented standards for data collection and reporting.	_____
D5	The use of centralized BI tools has improved governance standardization in our PMO.	_____

Thank you for completing this survey. Your responses are anonymous and will be used solely for academic research purposes in partial fulfilment of the Bachelor's thesis requirements at Riga Nordic University.

B. Expert Evaluation Criteria for Final Thesis Quality

Table B.1. Expert evaluation criteria for the quality of the final thesis

No.	Criterion title and content	Points
1	Conformity of the thesis title to the research area; style; correctness of key words.	5
2	Thesis structure: presence of the essential chapters.	5
3	Level of research motivation grounded in the contemporary state of the problem.	5
4	Quality and correctness of the theses to be defended; novelty of the main results.	5
5	Adequacy of the research methodology to the formulated research problem; appropriateness of the theoretical framework.	5
6	Originality of the research problem.	5
7	Use of constructive methods (including mathematical and computational) in solving the research problems.	5
8	Quality of conclusions: justification and correctness.	5
9	Quality of thesis formatting, literary style, compliance with standards and regulations.	5
10	Quality of the list of references: currency, authority, presence of sources in foreign languages.	5

Note.

1. All criteria are mandatory.
2. Final thesis evaluation:
 - < 25 points — the thesis does not meet the requirements;
 - 25–35 points — the thesis requires improvement;
 - 36–50 points — the thesis is eligible for defense.

Apliecinājums / Affirmation

Ar šo es, Amina Showkath, apliecinu, ka bakalaura darbs ir izpildīts patstāvīgi, bez citu palīdzības, no svešiem avotiem ņemtie dati un definējumi ir uzrādīti darbā. Šis darbs nekādā veidā nav iesniegts nevienai citai pārbaudījuma komisijai un nekur nav publicēts.

(Hereby I, Amina Showkath, affirm that the Bachelor's thesis was performed independently; sources of data and definitions are provided. This work has not been submitted to any other examination commission and has not been published elsewhere.)

Rīga, 17.04.2026.

paraksts/signature, atšifrējums/decription

Pateicības / Acknowledgments

Autors izsaka pateicību darba vadītājam par sniegtajiem padomiem un atbalstu, kā arī visiem aptaujas dalībniekiem un ekspertam par veltīto laiku un ieguldījumu pētījuma realizēšanā.

The author expresses gratitude to the thesis supervisor for the provided advice and support, as well as to all survey participants and the interview expert for their time and valuable contribution to this research.